

28

A

$$P_{\text{mean}(p)} = V_{\text{rms}(p)} I_{\text{rms}(p)}$$

$$3600 = 500 I_{\text{rms}(p)}$$

$$I_{\text{rms}(p)} = 7.2 \text{ A}$$

$$\frac{I_{\text{rms}(p)}}{I_{\text{rms}(s)}} = \frac{N_s}{N_p}$$

$$\frac{7.2}{I_{\text{rms}(s)}} = \frac{480}{120}$$

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$$I_{\text{rms}(s)} = 1.8 \text{ A}$$

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$$P_{\text{mean}(cables)} = I_{\text{rms}(s)}^2 R = 1.8^2 (80) = 260 \text{ W}$$

